

Justin Lee

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EDUCATION

Rice University

Houston, TX

Bachelor of Science in Computer Science, Minor in Data Science

May 2028

- Relevant Coursework: Object Oriented Programming, Data Structures, Algorithms, Computer Architecture, Multivariable Calculus, Linear Algebra.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, SQL

Technologies: React, Next.js, FastAPI, Node.js, Pandas, NumPy, LangChain, Hugging Face

Databases: PostgreSQL, MongoDB, MySQL, LanceDB

EXPERIENCE

OpenStax

Houston, TX

AI Engineer

May 2026 – Present

- Led 5 engineers in development of an interactive branching-scenario platform from schema design to deployed UI, generating 50+ nursing education scenarios.
- Built an automated ingestion pipeline (BeautifulSoup, markdownify) that parsed 5 OpenStax textbooks (1,000+ pages each) into structured Markdown, cutting content prep from 100+ hours to minutes.
- Designed the system to run on either Claude or Gemini interchangeably and tuned prompts so generated content followed formatting standards, cutting out-of-spec outputs by 80%.
- Architected a Next.js + FastAPI + PostgreSQL stack modeling narrative logic as a directed graph, with React components rendering and editing AI-generated JSON nodes in real time.

Rice University Digital Health Initiative

Houston, TX

Software Engineer

November 2025 – March 2026

- Built an MVP marketplace matching undergraduate, graduate, and PhD students to medical research projects with layered eligibility, skill tags, and relevance ranking.
- Designed a structured intake that turns brief project inputs into conversion-optimized opportunity listings with clear scope, contact, and timelines, improving applicant quality.
- Implemented role-based access, candidate profiles, and fast search/filter pipelines to surface both short- and long-term projects and accelerate staffing across labs.
- Partnered with a Rice professor and Houston Methodist stakeholders to define the roadmap; MVP launched with a committed client pipeline providing real users and immediate adoption signals.

PROJECTS

OwlConnect | *LangChain, FastAPI, MongoDB, Next.js, WebSocket, MCP*

September 2025

- Built an AI mentorship platform for Rice students, using All-MiniLM-L6-v2 vector embeddings with a LangChain multi-agent negotiation system to boost match accuracy.
- Engineered a FastAPI + MongoDB vector DB backend with MCP pipelines aligning academic and professional data.
- Developed a Next.js + TypeScript frontend with WebSocket-driven visualizations, mapping mentor expertise to career trajectories in real time.
- Won 2nd Place Overall out of 100+ teams at HackRice, Rice University's annual hackathon.

HootPlanner | *React, TypeScript, FastAPI, Pandas, LanceDB*

September 2024

- Developed an AI-powered academic course recommendation platform with a React/TypeScript frontend and FastAPI backend.
- Implemented an ETL pipeline using Pandas for preprocessing, all-MiniLM-L12-v2 for 384-dimensional embeddings, and LanceDB to store vectors for fast semantic search.
- Won 1st Place Overall out of 100+ teams at HackRice, Rice University's annual hackathon.